Retail Sales Analysis and Prediction Using E-Commerce Data

DSAI4103 Business Analytics

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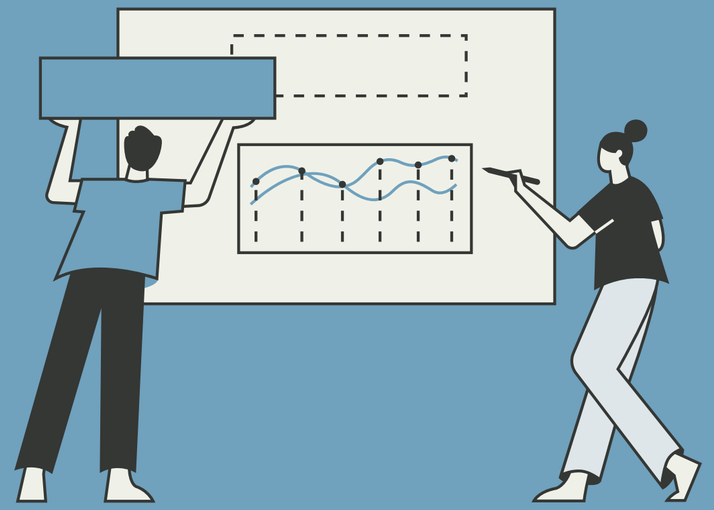
Business Problem

E-commerce companies generate large amounts of data but struggle to extract useful insights.

This project analyzes sales data to support better business decision-making.

Objectives

1. Analyze sales trends over time
2. Identify top product categories
3. Understand customer behavior
4. Predict order value using machine learning



Dataset

Brazilian E-commerce dataset (Olist)

Includes orders, customers, products, payments, reviews

~100,000 records

Data Processing

Merged multiple datasets

Handled missing values

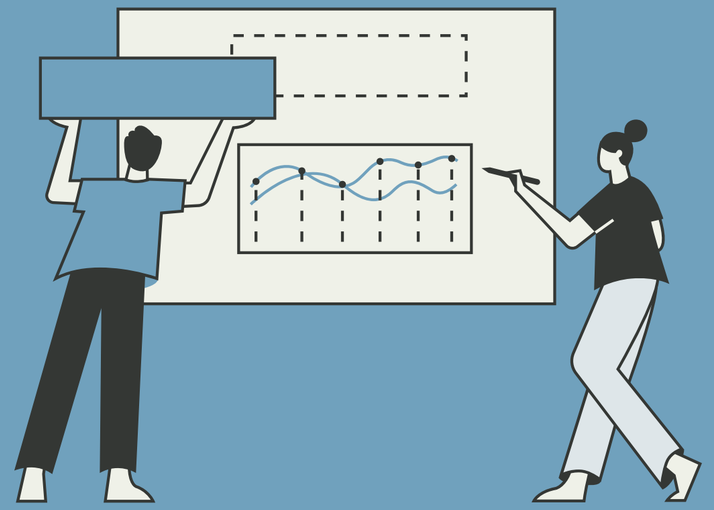
Cleaned and prepared data

Exploratory Data Analysis

1. Sales trends analyzed over time
2. Top product categories identified
3. Customer distribution explored

Model

1. Random Forest model used
2. Predicted order value
3. Evaluated using MAE and R^2
4. Model deployed with scoring output



Model Results

MAE ≈ 108

Model shows moderate prediction error

R^2 Score is Low

Results integrated into dashboard for evaluation

Explainability

Used feature importance

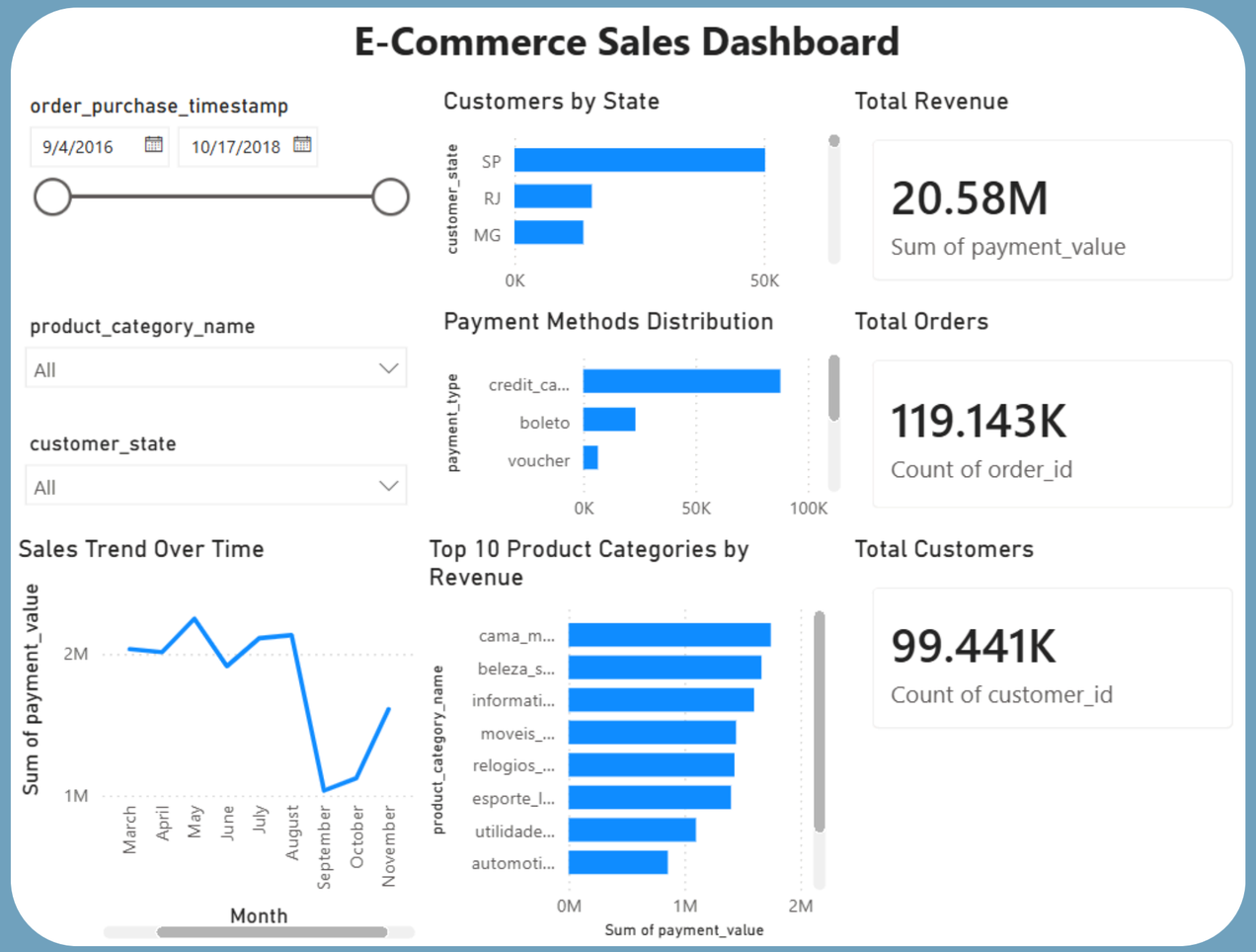
Price and freight value are key factors

Product size has moderate impact

Bias Analysis

1. Compared customer groups (state, payment type)
2. Some variation observed
3. No significant bias detected

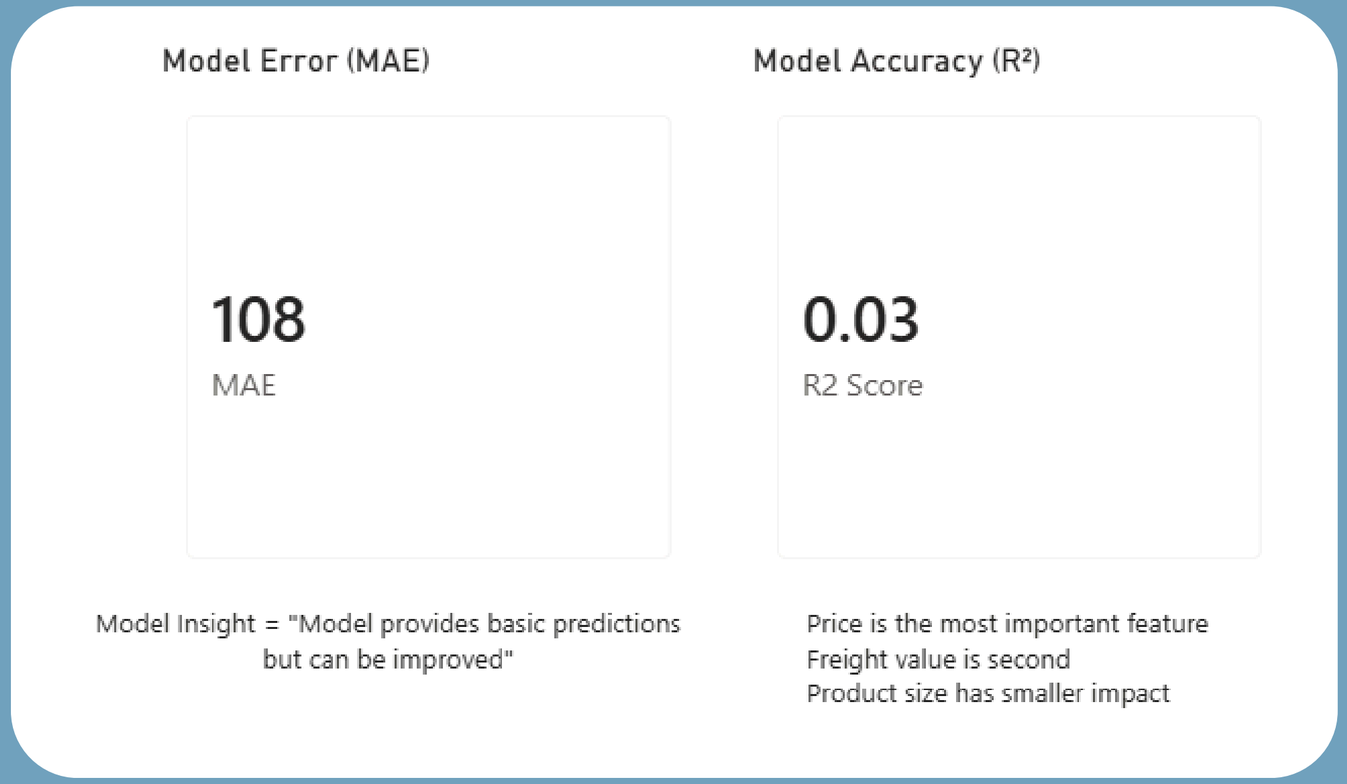
Power BI Dashboard



Sales Overview Page

- Multi-page interactive Power BI dashboard
- Provides sales, customer, and product insights
- Includes model performance (scoring board)
- Interactive filters synchronized across all pages

Model Performance Dashboard



Model Performance Page

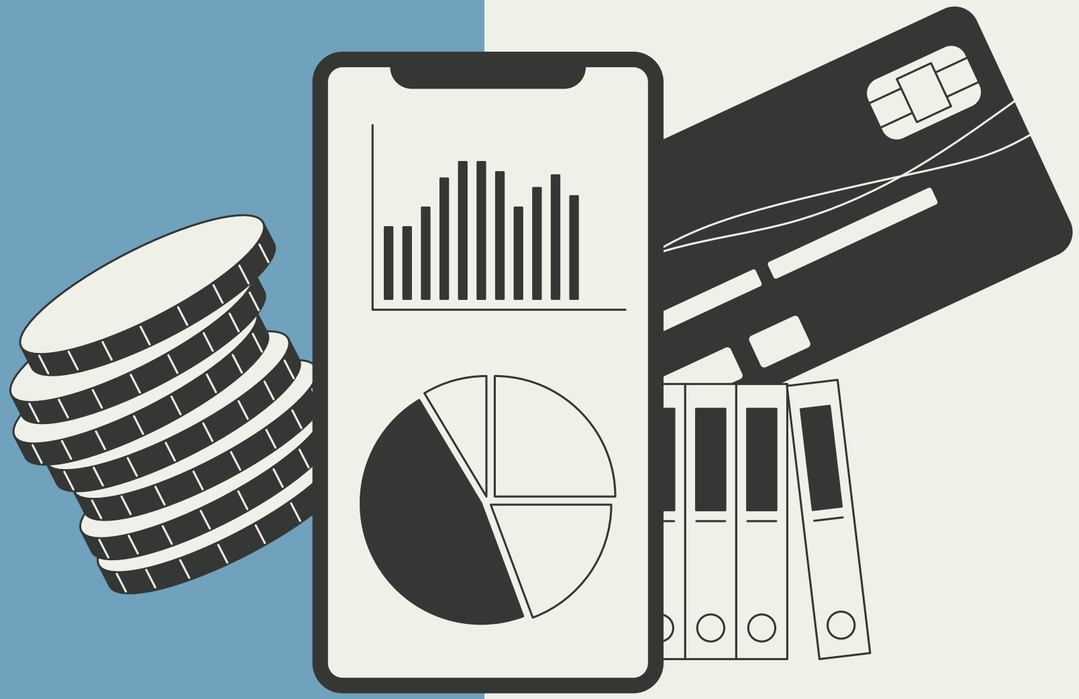
- MAE and R^2 displayed as KPIs
- Model performance visualized
- Helps evaluate model predictions
- Provides clear model performance summary

Conclusion

Sales driven by price and shipping cost

Some product categories perform better

Dashboard supports data-driven decisions

Thank You

Thank you for your attention